

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Solubility

1. Which of the following halide is soluble in water?

(A) AgF (B) AgCl (C) AgBr (D) AgI

CQA160

Test of Carbonate and Bicarbonate

2. A gas turns lime water milky and acidified $K_2Cr_2O_7$ solution green then gas is:

(A) HCl (B) H_2S (C) SO_2 (D) CO_2

CQA155

Test of Sulphite (SO_3^{2-}); Bisulphite (HSO_3^-)

3. When a salt is heated with dilute H_2SO_4 and $KMnO_4$ solution, the pink colour of $KMnO_4$ is discharged, the salt is :

(A) a sulphite (B) a carbonate (C) a nitrate (D) a bicarbonate

CQA145

Test of Sulphide (S^{2-})

4. $Pb(CH_3COO)_2$ gives.....colour with H_2S :-

(A) Black (B) White (C) Red (D) Orange

CQA190

Test of Acetate (CH_3COO^-)

5. When CH_3COONa heated with solid As_2O_3 then compound X is formed. The smell of compound X is

(A) Pungent smell (B) Rotten Fish smell (C) Nausting smell (D) Rotten egg smell

CQA094

Test of Nitrite (NO_2^-)

6. A gas has smell like rotten egg and turns lead acetate paper black. The gas is :

(A) NO_2 (B) H_2S (C) CO_2 (D) SO_2

CQA156

7. When KI is added to acidified solution of sodium nitrite :

(A) NO gas is liberated and I_2 is set free (B) N_2 gas is liberated and HI is produced
(C) N_2O gas is liberated and I_2 is set free (D) N_2 gas is liberated and HOI is produced

CQA159

"Wet test of Acidic Radicals of Class A subgroup-II (Preliminary Tests)"

8. Which of the following anion behaves in a different manner than other on heating with conc. H_2SO_4 ?

(A) Cl^- (B) I^-
(C) Br^- (D) All behave in a similar manner

CQA151

Test of Chloride (Cl^-), Bromide (Br^-), Iodide (I^-)

9. When $K_2Cr_2O_7$ is heated with conc. H_2SO_4 and soluble chloride such as KCl :

(A) red vapours of CrO_2Cl_2 are evolved (B) Cl^- ion is oxidized to Cl_2 gas
(C) $CrCl_3$ is formed (D) $Cr_2O_7^{2-}$ ion is reduced to green Cr^{3+} ion

CQA148

10. Which of the following will not give positive chromyl chloride test ?
 (A) Copper chloride, CuCl_2 . (B) Mercuric chloride, HgCl_2 .
 (C) Zinc chloride, ZnCl_2 . (D) Anilinium chloride $\text{C}_6\text{H}_5\text{NH}_3\text{Cl}$.

CQA152

Test of Nitrate (NO_3^-)

11. The brown ring test for NO_2^- and NO_3^- is due to formation of complex ion with formula :
 (A) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (B) $[\text{Fe}(\text{CN})_5(\text{NO})]^{2-}$ (C) $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$ (D) $[\text{Fe}(\text{H}_2\text{O})(\text{NO})_5]^{2+}$
 12. Comment True (T) or False (F) for the following statements.
 (I) Same gas comes out when Na_2S is treated with dil. HCl and Na_2SO_3 is treated with $(\text{Zn} + \text{dil. H}_2\text{SO}_4)$
 (II) Same observation when acidified nitrite solution is treated with urea or thiourea followed by addition of FeCl_3 solution.
 (III) Same observation when nitrite solution or acetate solution is treated with dil. H_2SO_4 solution
 (IV) With CaCl_2 and BaCl_2 both acetate and formate do not react.
 (A) FTTF (B) TFFT (C) TFFT (D) FTTT

CQA099

Test of Oxalate ($\text{C}_2\text{O}_4^{2-}$)

13. Which of the following pair of acidic radical can be distinguished by using dil H_2SO_4 ?
 (I) $\text{C}_2\text{O}_4^{2-}$ and NO_3^- (II) NO_3^- and NO_2^- (III) Cl^- and Br^- (IV) HCO_3^- and CO_3^{2-}
 (A) I and II (B) II only (C) II and IV (D) III and IV
 14. Salt 'X' + dil. $\text{H}_2\text{SO}_4 \longrightarrow (\text{B})_{\text{gas}} + \text{H}_2\text{O}$
 Salt 'Y' + conc. $\text{H}_2\text{SO}_4 \longrightarrow (\text{B})_{\text{gas}} + (\text{C})_{\text{gas}} + \text{H}_2\text{O}$
 Then salt 'X' and 'Y' may contain anions respectively :
 (A) $\text{SO}_3^{2-}, \text{S}^{2-}$ (B) $\text{C}_2\text{O}_4^{2-}, \text{CO}_3^{2-}$ (C) $\text{CO}_3^{2-}, \text{C}_2\text{O}_4^{2-}$ (D) $\text{SO}_3^{2-}, \text{S}_2\text{O}_3^{2-}$

CQA150

CQA191

Class (B) : Test of Phosphate (PO_4^{3-})

15. A metal salt solution gives a yellow precipitate with silver nitrate. The precipitate dissolves in dil. Nitric acid as well as in ammonium hydroxide. The solution contains.
 (A) Br^- (B) I^- (C) PO_4^{3-} (D) SO_4^{2-}

CQA140

Classification of Basic Radicals

16. Identify the **INCORRECT** group reagent for their respective cations .
 (A) $\text{Pb}^{2+}, \text{Cu}^{2+}, \text{As}^{3+}$, – dil. $\text{HCl} + \text{H}_2\text{S}$
 (B) $\text{Al}^{3+}, \text{Fe}^{3+}, \text{Cr}^{3+}$, – $\text{NH}_4\text{Cl} + \text{NH}_4\text{OH}$
 (C) $\text{Zn}^{2+}, \text{Ni}^{2+}, \text{Mn}^{2+}, \text{Co}^{2+}$ – $\text{NH}_4\text{Cl} + \text{NH}_4\text{OH} + \text{H}_2\text{S}$
 (D) $\text{Ba}^{2+}, \text{Cr}^{2+}, \text{Ca}^{2+}$, – dil. HCl

CQA192

Test of Basic Radicals: Group Zero

17. Ammonium salt gives brown colour with alkaline solution of Nessler's reagent to form iodide of Millon's base. The formula of this compound is :-

(A) $\text{NH}_2\text{-Hg-OHg-I}$ (B) $\text{NH}_2\text{-O-Hg-HgI}$
(C) K_2HgI_4 (D) $\text{NH}_2\text{-Hg-I}$

CQA193
Test of Basic Radicals: Group 1

18. A cold aqueous solution of PbCl_2 gives golden yellow precipitate on addition of
(A) KCl solution (B) KI solution (C) NaCl solution (D) K_2SO_4 solution

CQA142
Test of Basic Radicals: Group 2A

19. Which of the following reagents turns white precipitate of AgCl yellow?
(A) NaNO_3 (B) Na_3AsO_3 (C) Na_3AsO_4 (D) NaCN
20. When a metal sulphate is heated in dry test tube, the colour changes from blue to white. Then metal sulphate may be :
(A) BaSO_4 (B) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (C) Na_2SO_4 (D) None of these

CQA139
CQA153
Test of Basic Radicals: Group 2B

21. In which of the following pairs the precipitates are red and black coloured respectively and both precipitates are soluble in excess KI solution ?
(A) $\text{HgI}_2, \text{Hg}_2\text{I}_2$ (B) $\text{HgI}_2, \text{BiI}_3$ (C) $\text{Cu}_2\text{I}_2, \text{AgI}$ (D) $\text{CdI}_2, \text{PbI}_2$
22. A 0.1 M solution of certain cation will precipitate with 0.1 M solutions of all these anions $\text{OH}^\ominus$, $\text{CO}_3^{2\ominus}$ & $\text{SO}_4^{2\ominus}$. This description is true for the cation.
(A) $\text{Fe}^{2\oplus}$ (B) $\text{Mg}^{2\oplus}$ (C) $\text{Ba}^{2\oplus}$ (D) $\text{Pb}^{2\oplus}$
23. When bismuth chloride is poured into a large volume of water, the white precipitate produced is:
(A) Bi(OH)_3 (B) Bi_2O_3 (C) BiOCl (D) Bi_2OCl_3

CQA123
CQA158
CQA194
Test of Basic Radicals : Group 3

24. Which of the following combines with Fe(II) ions to form a brown complex?
(A) N_2O (B) NO (C) N_2O_5 (D) N_2O_4
25. Sulphide ion reacts with $\text{Na}_2[\text{Fe(CN)}_5\text{NO}]$ to form a purple coloured compound (X). In this reaction oxidation state of iron.
(A) changes from +2 to +3 (B) changes from +3 to +2
(C) changes from +2 to +4 (D) does not change.
26. Fe(OH)_3 and Cr(OH)_3 precipitates can be completely separated by :
(A) Aq. NH_3 (B) HCl (C) NaOH/ H_2O_2 (D) H_2SO_4

CQA136
CQA144
CQA119

27. An original solution of an inorganic salt in dilute HCl gives a brown colouration with potassium hexacyanidoferrate (III) and reddish-brown colouration with sodium acetate solution. The cation of the salt is :

(A) $\text{Ni}^{2\oplus}$ (B) $\text{Fe}^{3\oplus}$ (C) $\text{Cu}^{2\oplus}$ (D) none

CQA127

28. An inorganic lewis acid (X) fumes in moist air, and intensity of fumes increases when rod dipped in NH_4OH is brought near to it. An acidic solution of (X) on addition of NH_4Cl and NH_4OH gives a precipitate which dissolves in NaOH solution. A solution of (X) does not give precipitate with H_2S Hence, compound (X) is

(A) FeCl_3 (B) AlCl_3 (C) CrCl_3 (D) ZnCl_2

CQA126

29. Potassium thiocyanate solution reacts with ferric chloride to give :-

(A) Pink colour (B) Deep blue colour
(C) Green colour (D) Blood-red colour

CQA195

30. Fe is precipitated as $\text{Fe}(\text{OH})_3$ and not $\text{Fe}(\text{OH})_2$ because :-

(A) $\text{Fe}(\text{OH})_2$ is not precipitated as it is soluble
(B) $\text{Fe}(\text{OH})_3$ is soluble
(C) $\text{Fe}(\text{OH})_2$ is insoluble
(D) None of these

CQA196

31. $\text{Fe}_2(\text{SO}_4)_3$ solution + $\text{Na}_2\text{S}_2\text{O}_3$ solution.

Select correct about above reaction.

(A) Initially violet solution produces (B) Finally redox change is observed
(C) $\text{S}_2\text{O}_3^{2\ominus}$ is oxidised to $\text{S}_2\text{O}_6^{2\ominus}$ (D) All of the above

CQA197

32. $\text{Fe}^{3\oplus}$ ions produce deep blue colouration when treated with reagent (ft). R is :

(A) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (B) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (C) KSCN (D) excess of NH_4OH

CQA198

Test of Basic Radicals : Group 4



Colour of precipitate of "X" is :

(A) green (B) black (C) yellow (D) Reddish Brown

CQA143

34. H_2S in the presence of HCl precipitates IInd group but not IV group because :

(A) HCl activates H_2S
(B) HCl increases concentration of $\text{Cl}^\ominus$
(C) HCl decreases concentration of $\text{S}^{2\ominus}$
(D) HCl lowers the solubility of H_2S in solution

CQA120

35. A coloured solution of a salt gives following reactions.
 (i) It gives white precipitate with sodium hydroxide which becomes brown on exposure to air.
 (ii) It gives light pink precipitate with NH_4OH solution which is insoluble in excess NH_3 .
 (A) Mn^{2+} (B) Zn^{2+}
 (C) Al^{3+} (D) Ni^{2+} CQA128
36. Select compound which is written with its colour correctly.
 (A) K_2MnO_4 (Purple) (B) CdS black
 (C) Cr_2O_3 , (Green) (D) MnS (Black) CQA199
37. Which pair of ions can not be distinguished by given reagent?

Pair of ions	Reagent
(A) Fe^{2+} , Fe^{3+} $\Rightarrow$	$\text{K}_4[\text{Fe}(\text{CN})_6](\text{aq.})$
(B) CO_3^{2-} , HCO_3^- $\Rightarrow$	CaCl_2 (aq.)
(C) Hg^{2+} , Ag^+ $\Rightarrow$	HCl (dil.)
(D) Al^{3+} , Zn^{2+} $\Rightarrow$	NaOH (excess)

CQA200

Test of Basic Radicals : Group 5

38. $\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \longrightarrow \text{White precipitate}$ $\xrightarrow[\text{substance with H}_2\text{O}]{\text{Fuse this precipitate on charcoal with Na}_2\text{CO}_3 \text{ and extract the soluble}}$ Aqueous solution. $\xrightarrow[\text{heat the solution}]{\text{Add dil H}_2\text{SO}_4 \text{ and}}$ Gas 'G' is evolved.
 The gas 'G' will show which of the following property ?
 (A) It turns lead acetate filter paper black.
 (B) It turns acidified $\text{K}_2\text{Cr}_2\text{O}_7$ filter paper green.
 (C) It produces purple colouration on filter paper moistened with sodium nitroprusside already made alkaline with sodium hydroxide.
 (D) All of these CQA157
39. There are four test tubes containing dilute HCl , BaCl_2 , HgCl_2 and KNO_3 solutions. Which of the following reagents will help in the identification of BaCl_2 ?
 (A) NaOH (B) K_2CrO_4 (C) AgNO_3 (D) both (B) and (C) CQA147
40. A salt on treatment with dil. HCl gives a pungent smelling gas and a yellow precipitate. The salt gives green flame when tested. The solution gives a yellow precipitate with potassium chromate. The salt is:
 (A) NiSO_4 (B) BaS_2O_3 (C) PbS_2O_3 (D) CuSO_4 CQA202

Dry tests for Basic Radicals: Flame test

41. Metal (M) shows crimson red colour in flame test and its halide is deliquescent then metal (M) could be :

(A) Li (B) Mg (C) Ca (D) Ba

CQA138

42. A fire work gave bright crimson red light. It probably contained a salt of :

(A) Ca (B) Sr (C) Ba (D) Mg

CQA141

Borax Bead test, Microcosmic Bead Test

43. Which one of the following ions does not give borax bead test ?

(A) Cr^{3+} (B) Cu^{2+} (C) Mn^{2+} (D) Zn^{2+}

CQA137

44. In the borax bead test of Co^{2+} , the blue colour of bead is due to the formation of:

(A) B_2O_3 (B) Co_3B_2 (C) $\text{Co}(\text{BO}_2)_2$ (D) CoO

CQA111

Heating Effect

45. Which of the following can not evolve more than one gas (vapour) if heated in dry test tube.

(A) $\text{NaNO}_3(\text{s})$ (B) $\text{MgCO}_3(\text{s})$ (C) $\text{FeSO}_4(\text{s})$ (D) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7(\text{s})$

CQA154

46. Which of the following leaves no residue on heating?

(A) $\text{Pb}(\text{NO}_3)_2$ (B) NH_4NO_3 (C) $\text{Cu}(\text{NO}_3)_2$ (D) NaNO_3

CQA112

EXERCISE - (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Test of Sulphite (SO_3^{2-}); Bisulphite (HSO_3^-)

1. H_2S and SO_2 can be distinguished by :
 (A) Litmus paper (B) MnO_4^-/H^+ (C) $(CH_3COO)_2Pb$ (D) None of these

CQA164

Test of Basic Radicals : Group 1

2. Ion(s) which do(es) not produce ppt. with excess of CN^- ions :
 (A) K^+ (B) Pb^{2+} (C) Hg_2^{2+} (D) Ag^+

CQA203

Test of Basic Radicals: Group 2B

3. Which of the following is/are correctly matched ?
 (A) $BiI_3 \downarrow \longrightarrow$ Black precipitate (B) $CuI \downarrow \longrightarrow$ White precipitate
 (C) $PbI_2 \downarrow \longrightarrow$ Yellow precipitate (D) $HgI_2 \downarrow \longrightarrow$ Red precipitate
4. Which of the following statement(s) is/are correct ?
 (A) Yellow precipitated of silver arsenite is soluble in both nitric acid and ammonia.
 (B) Potassium cyanide when added in very small quantity to copper sulphate solution, produces first yellow precipitate which quickly converts in to white precipitate.
 (C) Black precipitate of BiI_3 turns orange on heating with water.
 (D) White precipitate of $Bi(OH)_3$ turns yellowish brown, when boiled.
5. Which of the following statements is/are incorrect ?
 (A) A filter paper moistened with cadmium acetate solution turns yellow, when brought in contact with H_2S gas.
 (B) Both carbonate ions as well as bicarbonate ions in the solutions, give reddish-brown precipitate with mercury(II) chloride.
 (C) Sulphites in presence of zinc, reacts with dilute H_2SO_4 to liberate SO_3 gas.
 (D) None of these

CQA161

CQA163

CQA165

6. Which anion can produce ppt. with lead acetate solution?
 (A) NO_3^- (B) Cl^- (C) CO_3^{2-} (D) S^{2-}

CQA204

Test of Basic Radicals : Group 4

7. $Co^{2+} + KCN$ (not in excess) $\longrightarrow$ precipitate.
 Select the correct statement(s) with respect to the precipitate.
 (A) It is yellow in colour.
 (B) It is reddish-brown in colour.
 (C) It dissolves in excess of the reagent forming a brown solution.
 (D) It is obtained when brown solution (option (C)) is acidified with dilute HCl in the cold.

CQA162

8. Which of the following combination will give a red colour or precipitate in an aqueous medium?
 (A) $\text{Fe}^{3+} + \text{SCN}^-$ (B) $\text{Fe}^{3+} + [\text{Fe}(\text{SN})_6]^{3-}$
 (C) $\text{Ni}^{2+} + \text{dimethyl glyoxime}$ (D) $\text{Co}^{2+} + \text{SCN}^-$ CQA134
9. $\text{CoCl}_2 + \text{KNO}_2 + \text{CH}_3\text{COOH} \longrightarrow [\text{X}] + \text{H}_2\text{O} + \text{KCl} + \text{CH}_3\text{COOK} + \text{NO}$
 (Unbalanced equation)
 Which is / are correct for 'X'?
 (A) X is a yellow crystalline solid which is insoluble in water.
 (B) X is a green coloured compounds known as Rinman's green.
 (C) IUPAC name of X is potassium hexanitrito -N- cobaltate (II)
 (D) The compound X is an inner orbital complex. CQA205

Test of Basic Radicals : Group 5

10. KCN is used for separating :
 (A) Co^{2+} and Ni^{2+} (B) Mn^{2+} and Zn^{2+} (C) Ba^{2+} and Ca^{2+} (D) Cu^{2+} and Cd^{2+} CQA133

Borax Bead test, Microcosmic Bead Test

11. Which of the following can be detected by the borax bead test ?
 (A) Ni^{2+} (B) Co^{2+} (C) Pb^+ (D) Hg_2^{2+} CQA121

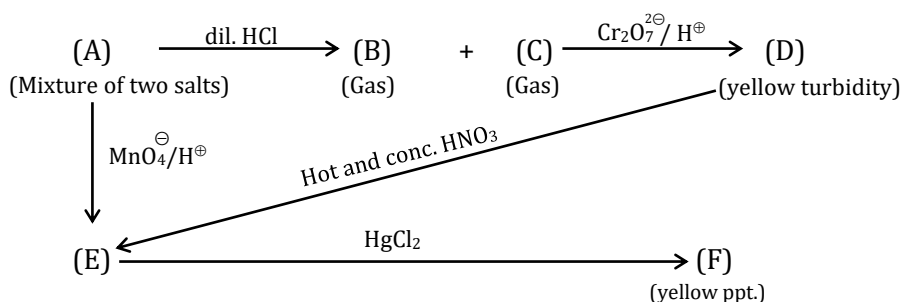
Heating Effect

12. Which of the following metal carbonate will give of metal and oxygen on heating-
 (A) Ag_2CO_3 (B) HgCO_3 (C) $(\text{NH}_4)_2\text{CO}_3$ (D) PbCO_3 CQA097
13. Which of the following metal nitrate produce NO_2 on heating
 (A) $\text{Hg}(\text{NO}_3)_2$ (B) RbNO_3 (C) $\text{Pb}(\text{NO}_3)_2$ (D) $\text{Cu}(\text{NO}_3)_2$ CQA110
14. Which of the following salts release reddish brown gas when heated in a dry test tube?
 (A) LiNO_3 (B) KNO_3 (C) $\text{Pb}(\text{NO}_3)_2$ (D) AgNO_3 CQA166
15. Which of the following pairs produce same gas?
 (A) $(\text{Ca}_3\text{N}_2 + \text{dil. HCl})$ and $(\text{NH}_4\text{NO}_2 \text{ on heating})$
 (B) $[(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \text{ on heating}]$ and $(\text{NH}_4\text{NO}_2 \text{ on heating})$
 (C) $(\text{NH}_4\text{NO}_3 \text{ on heating})$ and $[\text{Hg}(\text{NO}_3)_2 \text{ on heating}]$
 (D) $(\text{NH}_4\text{Cl} \text{ on heating})$ and $[\text{NaNO}_3 + \text{Zn} + \text{NaOH}_{(\text{aq.})} \text{ on heating}]$ CQA206

COMPREHENSION TYPE QUESTIONS

Class (B) : Test of Sulphate (SO_4^{2-})

Paragraph for Question No. 16 to 18



16. Find the anion(s) :
 (A) $\text{SO}_3^{2\ominus}$ (B) $\text{SO}_3^{2\ominus}, \text{S}^{2\ominus}$ (C) $\text{SO}_3^{2\ominus}, \text{SO}_4^{2\ominus}$ (D) $\text{S}_2\text{O}_3^{2\ominus}$ CQA173
17. Find out (E) :
 (A) $\text{S}^{2\ominus}$ (B) $\text{CO}_3^{2\ominus}$ (C) $\text{S}_2\text{O}_3^{2\ominus}$ (D) $\text{SO}_4^{2\ominus}$ CQA174
18. Find out (F) :
 (A) $\text{HgSO}_4 \cdot 2\text{HgO}$ (B) $\text{HgSO}_4 \cdot 3\text{HgO}$ (C) HgSO_4 (D) $\text{Hg}_2\text{SO}_4 \cdot 3\text{HgO}$ CQA175

Test of Basic Radicals : Group 4

Paragraph for Question No. 19 to 20

Two students were given mixture of $\text{Al}^{3\oplus}$, $\text{Cr}^{3\oplus}$, $\text{Fe}^{3\oplus}$, $\text{Zn}^{2\oplus}$, $\text{Ni}^{2\oplus}$, $\text{Co}^{2\oplus}$ and $\text{Mn}^{2\oplus}$ ions as their soluble salt which were not interfering radicals. They performed the experiment as follows:

Student I: In the aqueous solution of ions he added first NH_4Cl then NH_4OH and then passed H_2S gas. He got the precipitate (X).

student II: In the aqueous solution of ions he first added NH_4Cl and then NH_4OH . He got the precipitate (Y) and a filtrate. In the filtrate, he passed H_2S gas and again got a precipitate (Z). Based on above information answer the following two questions:

19. Which of the following is not the precipitate (X) ?
 (A) ZnS (B) MnS (C) $\text{Cr}(\text{OH})_3$ (D) Al_2S_3 CQA207
20. Precipitate (Y) when treated with aqueous Na_2O_2 solution, gives a yellow colour solution due to formation of which of the following?
 (A) ZnO (B) Na_2CrO_4 (C) $\text{Na}_2\text{Cr}_2\text{O}_4$ (D) $\text{Fe}(\text{OH})_3$ CQA208

Test of Basic Radicals : Group 5

Paragraph For Questions Q.21, Q.22 and Q.23 by appropriately matching the information given in the three columns of the following table.

Column-1		Column-2		Column-3	
(I)	$\text{Bi}^{3\oplus}$	(i)	NH_3	(P)	Black / white precipitate is obtained.
(II)	$\text{Cu}^{2\oplus}$	(ii)	KI	(Q)	Blue / Brown precipitate is obtained.
(III)	$\text{Fe}^{3\oplus}$	(iii)	KCN	(R)	Form soluble complex in excess reagent.
(IV)	$\text{Zn}^{2\oplus}$	(iv)	$\text{K}_4[\text{Fe}(\text{CN})_6]$	(S)	Redox reaction take place.

21. The only incorrect combination is:
 (A) (I) (ii) (P) (B) (II) (ii) (S) (C) (II) (ii) (R) (D) (III) (iii) (R) CQA167
22. The only correct combination is:
 (A) (II) (ii) (R) (B) (III) (ii) (Q) (C) (III) (i) (R) (D) (II) (iii) (S) CQA168
23. The only incorrect combination is:
 (A) (III) (iv) (Q) (B) (IV) (ii) (P) (C) (I) (i) (P) (D) (IV) (i) (R) CQA169

MATRIX MATCH TYPE QUESTION

Test of Sulphite (SO_3^{2-}); Bisulphite (HSO_3^-)

24. Match List-I with List-II and select the correct answer using the codes given below the lists :

	List-I		List-II
P.	White turbidity	1.	$\text{IO}_3^- + \text{SO}_2 + \text{starch} \longrightarrow$
Q.	Rotten egg smell	2.	$\text{SO}_2 + \text{MnO}_4^- \longrightarrow$
R.	Colourless solution	3.	$\text{Zn} + \text{NaOH} + \text{SO}_2 \longrightarrow$
S.	Blue colour	4.	$\text{CO}_2 + \text{Ca(OH)}_2 \longrightarrow$

Code :

	P	Q	R	S
(A)	1	3	2	4
(B)	3	2	4	1
(C)	4	3	2	1
(D)	4	1	2	3

CQA176

Test of Basic Radicals : Group 5

25. Match list

	List-I (Basic Radical)		List-II (Group)
P	Al^{3+}	1	II group
Q	Zn^{2+}	2	V group
R	Ba^{2+}	3	IV group
S	Pb^{2+}	4	III group

(A) P-4; Q-2; R-1; S-3 (B) P-2; Q-4; R-1; S-3 (C) P-3; Q-1; R-2; S-4 (D) P-4; Q-3; R-2; S-1

CQA115

Test of Basic Radicals : Group 6

26. Match the column -

Column-I

(Radical which can decolorize acidic solution of MnO_4^-)

- (A) Only acidic radical
 (B) Only basic radical
 (C) both radical
 (D) neither acidic nor basic radical

Column-II

(Given salt)

- (P) $\text{Sn(NO}_3)_2$
 (Q) KNO_2
 (R) FeC_2O_4
 (S) Na_2CO_3
 (T) BaSO_3

- (A) (A) $\rightarrow$ (Q); (B) $\rightarrow$ (R); (C) $\rightarrow$ (R); (D) $\rightarrow$ (T)
 (B) (A) $\rightarrow$ (Q,T); (B) $\rightarrow$ (P); (C) $\rightarrow$ (P); (D) $\rightarrow$ (S)
 (C) (A) $\rightarrow$ (Q,T); (B) $\rightarrow$ (P); (C) $\rightarrow$ (R); (D) $\rightarrow$ (S)
 (D) (A) $\rightarrow$ (Q); (B) $\rightarrow$ (P); (C) $\rightarrow$ (S); (D) $\rightarrow$ (T)

CQA114

EXERCISE - (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Test of Sulphide (S^{2-})

1. $\text{Na}_2\text{S} + \text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}] \rightarrow \text{"A"}$ (Violet Color)
In Complex "A", number of type of ambidentate ligand is/are "a" and number of d-orbital involved in hybridisation is/are "b" Then $7a + 8b$ will be :

CQA185

Test of Chloride (Cl^-), Bromide (Br^-), Iodide (I^-)

2. $\text{NaCl} + \text{Solid K}_2\text{Cr}_2\text{O}_7 + \text{Conc. H}_2\text{SO}_4 \longrightarrow \text{"X"}$ (reddish brown fumes)
How many axial-d-orbital are involved in hybridization of "X" ?

CQA187

Test of Nitrate (NO_3^-)

3. Find the total number of acidic radical which produce volatile product with dil HCl :
- | | | | |
|------------------------|-----------------------|-----------------------|----------------------|
| (i) SO_4^{2-} | (ii) I^- | (iii) NO_2^- | (iv) NO_3^- |
| (v) SO_3^{2-} | (vi) HCO_3^- | | |

CQA182

4. How many of the following liberate coloured vapour/gas with concentrated H_2SO_4 ?
- $\text{KCl (s)} + \text{K}_2\text{Cr}_2\text{O}_7 \text{ (s)}, \text{KNO}_2 \text{ (s)}, \text{KI (s)}, \text{KBr (s)}, \text{KCl (s)}$
 $\text{KBr (s)} + \text{MnO}_2 \text{ (s)}, \text{KNO}_3, \text{KCl (s)} + \text{MnO}_2, \text{K}_2\text{SO}_3$

CQA186

Test of Borate (BO_3^{3-})

5. $\text{BO}_3^{3-} + \text{cone. H}_2\text{SO}_4 + \text{CH}_3 - \text{CH}_2 - \text{OH} \xrightarrow{\text{ignite}} \text{(A)}$

Green flame

What is the oxidation number of central atom that is responsible for green flame in compound (A)?

CQA184

Test of Oxalate ($\text{C}_2\text{O}_4^{2-}$)

6. $\text{Na}_2\text{CrO}_4 + \text{dil. H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{CrO}_4$
 $\text{H}_2\text{CrO}_4 + 2\text{H}_2\text{O}_2 \xrightarrow{\text{amyl alcohol}} \text{"A"} + \text{H}_2\text{O}$
 then find a + b, if
 a = number of O - O bonds in "A"
 b = number of Cr - O σ (sigma) bonds in "A"

CQA183

Test of Basic Radicals : Group 1

7. An aqueous solution contains Hg^{2+} , Hg_2^{2+} , Pb^{2+} , Ag^+ , Bi^{3+} and Cd^{2+} . Out of these, how many ions will produce white precipitate with dilute HCl ?

CQA210

Test of Basic Radicals: Group 2A

8. $\text{Cu}^{2+} + \text{KCN}$ (in excess) $\rightarrow$ soluble complex (X). How many statements are correct regarding complex (X) -
- (i) the central atom has the co-ordination number of 6
 - (ii) the central atom has the co-ordination number of 4
 - (iii) the complex is sq.planar
 - (iv) the complex is diamagnetic
 - (v) the complex is paramagnetic

CQA135

Test of Basic Radicals: Group 2B

9. Solution of AsO_4^{3-} ion containing considerable excess of HNO_3 on boiling with Ammonium molybdate test gives a yellow crystalline precipitate (X). Number of oxygen atom present in per molecule of precipitated (X) are :
10. Oxidation state of Hg in Nessler's Reagent is :-

CQA179

CQA211

Test of Basic Radicals : Group 3

11. $\text{Fe}^{2+}(\text{aq}) + \text{NO}_3^-(\text{aq}) + \text{H}_2\text{SO}_4(\text{conc}) \longrightarrow$ Brown ring.
The oxidation number of iron in brown ring complex is :
12. How many statements are correct ?
- (i) Like CO_3^{2-} , SO_3^{2-} also gives test with lime water.
 - (ii) In the test of NO_3^- , brown ring is formed due to the formation of $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$.
 - (iii) Aqueous solution of Ag^+ ions gives brick red/red precipitate with K_2CrO_4 .
 - (iv) Lead salts turn black on prolonged exposure in a chemical laboratory.
 - (v) In analysis of group IInd cations, H_2S gas is passed in presence of hydrochloric acid to enhance the concentration of S^{2-} ions.
 - (vi) Aqueous solution of Cu^{2+} ions forms a green complex with $\text{K}_4[\text{Fe}(\text{CN})_6]$.

CQA181

CQA188

Test of Basic Radicals : Group 4

13. An alcoholic solution of dimethylglyoxime is added to an aqueous solution of nickel(II) chloride. Slow addition of ammonium hydroxide led to the precipitation of a bright-red coloured metal complex.
Find out the number of hydrogen bonds present in the structure of the complex.
14. Which of the following sulphides are not black in colour.
 PbS , CdS , As_2S_3 , Sb_2S_3 , SnS , CoS , HgS , ZnS , MnS .

CQA178

CQA180

Heating Effect

15. On strong heating of H_3PO_4 and H_3BO_3 , sum of oxidation number of P & B in the final product obtained is

CQA122

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Test of Thiosulphate ($S_2O_3^{2-}$)

16. In neutral or faintly alkaline solution, 4 moles permanganate anion quantitatively oxidise thiosulphate anions to produce X moles of a sulphur containing product. The magnitude of $\frac{X}{9}$ is:

CQA212

Test of Chloride (Cl^-), Bromide (Br^-), Iodide (I^-)

17. $Fe^{2+} + NO_3^- + H_2SO_4(\text{conc.}) \rightarrow \text{'X'}$ (Brown ring complex)

The magnetic moment of complex 'X' :

CQA214

EXERCISE - JEE (Main) PYQ

1. Arrange the following in increasing order of solubility product : [JEE (Main) 2025]
 Ca(OH)_2 , AgBr , PbS , HgS
 (1) $\text{PbS} < \text{HgS} < \text{Ca(OH)}_2 < \text{AgBr}$ (2) $\text{HgS} < \text{PbS} < \text{AgBr} < \text{Ca(OH)}_2$
 (3) $\text{Ca(OH)}_2 < \text{AgBr} < \text{HgS} < \text{PbS}$ (4) $\text{HgS} < \text{AgBr} < \text{PbS} < \text{Ca(OH)}_2$
CQA223
2. What are the products formed in sequence when excess of CO_2 is passed in slaked lime? [JEE (Main) 2021]
 (1) $\text{Ca(HCO}_3)_2$, CaCO_3 (2) CaCO_3 , $\text{Ca(HCO}_3)_2$
 (3) CaO , $\text{Ca(HCO}_3)_2$ (4) CaO , CaCO_3
CQA048
3. Sodium extract is heated with concentrated HNO_3 before testing for halogens because : [JEE (Main) 2016]
 (1) S^{2-} and CN^- , if present are decomposed by conc. HNO_3 and hence do not interfere in the test.
 (2) Ag reacts faster with halides in acidic medium
 (3) Ag_2S and AgCN are soluble in acidic medium.
 (4) Silver halides are totally insoluble in nitric acid.
CQA041
4. A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the solution, a white precipitate is obtained which does not dissolve in dil. nitric acid. The anion is : [JEE (Main) 2018]
 (1) S^{2-} (2) SO_4^{2-} (3) CO_3^{2-} (4) Cl^-
CQA045
5. Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown - red vapours of: [JEE (Main) 2013]
 (1) CrO_3 (2) Cr_2O_3 (3) CrCl_3 (4) CrO_2Cl_2
CQA031
6. An organic compound 'A' is oxidized with Na_2O_2 followed by boiling with HNO_3 . The resultant solution is then treated with ammonium molybdate to yield a yellow precipitate. Based on above observation, the element present in the given compound is : [JEE (Main) 2019]
 (1) Sulphur (2) Nitrogen (3) Fluorine (4) Phosphorus
CQA047
7. On passing a gas, 'X', through Nessler's reagent, a brown precipitate is obtained. The gas 'X' is [JEE (Main) 2024]
 (1) H_2S (2) CO_2 (3) NH_3 (4) Cl_2
CQA215
8. Consider the following equilibrium [JEE (Main) 2013]
 $\text{AgCl} \downarrow + 2\text{NH}_3 \rightleftharpoons [\text{Ag}(\text{NH}_3)_2]^+ + \text{Cl}^-$
 White precipitate of AgCl appears on adding which of the following?
 (1) NH_3 (2) Aqueous NaCl
 (3) Aqueous NH_4Cl (4) Aqueous HNO_3
CQA035
9. Identify A, B and C in the given below reaction sequence [JEE (Main) 2025]
 $\boxed{\text{A}} \xrightarrow{\text{HNO}_3} \text{Pb(NO}_3)_2 \xrightarrow{\text{H}_2\text{SO}_4} \boxed{\text{B}} \xrightarrow[\text{(3) K}_2\text{CrO}_4]{\text{(1) Ammonium acetate, (2) Acetic acid}} \boxed{\text{C}}$
 Yellow ppt
 (1) PbCl_2 , PbSO_4 , PbCrO_4 (2) PbS , PbSO_4 , PbCrO_4
 (3) PbS , PbSO_4 , $\text{Pb(CH}_3\text{COO)}_2$ (4) PbCl_2 , $\text{Pb(SO}_4)_2$, PbCrO_4
CQA224

10. Identify incorrect statement [JEE (Main) 2013]
 (1) Copper (I) compounds are colourless except where colour results from charge transfer
 (2) Copper (I) compounds are diamagnetic
 (3) Cu_2S is black
 (4) Cu_2O is colourless

CQA034

11. Match List I with List II

List-I (Compound)		List-II (Colour)	
A.	$\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$	I.	Violet
B.	$[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$	II.	Blood Red
C.	$[\text{Fe}(\text{SCN})]^{2+}$	III.	Prussian Blue
D.	$(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$	IV.	Yellow

Choose the **correct** answer from the options given below :

[JEE (Main) 2024]

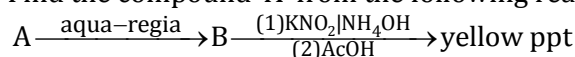
- (1) A-III, B-I, C-II, D-IV
 (2) A-IV, B-I, C-II, D-III
 (3) A-II, B-III, C-IV, D-I
 (4) A-I, B-II, C-III, D-IV

CQA217

12. An aqueous solution of a salt X turns blood red on treatment with SCN^- and blue on treatment with $\text{K}_4[\text{Fe}(\text{CN})_6]$. X also gives a positive chromyl chloride test. The salt X is : [JEE (Main) 2015]
 (1) CuCl_2 (2) $\text{Cu}(\text{NO}_3)_2$ (3) FeCl_3 (4) $\text{Fe}(\text{NO}_3)_3$

CQA037

13. Find the compound 'A' from the following reaction sequences. [JEE (Main) 2025]



- (1) ZnS (2) CoS (3) MnS (4) NiS

CQA225

14. A reagent which gives brilliant red precipitate with Nickel ions in basic medium is

[JEE (Main) 2024]

- (1) sodium nitroprusside (2) neutral FeCl_3
 (3) meta-dinitrobenzene (4) dimethyl glyoxime

CQA218

15. During the qualitative analysis of salt with cation Y^{2+} , addition of a reagent (X) to alkaline solution of the salt gives a bright red precipitate. The reagent (X) and the cation (Y^{2+}) present respectively are: [JEE (Main) 2022]

- (1) Dimethylglyoxime and Ni^{2+} (2) Dimethylglyoxime and CO^{2+}
 (3) Nessler's reagent and Hg^{2+} (4) Nessler's reagent and Ni^{2+}

CQA049

16. The **incorrect** statement is : [JEE (Main) 2018]

- (1) Cu^{2+} ion gives chocolate coloured precipitate with potassium ferrocyanide solution.
 (2) Cu^{2+} and Ni^{2+} ions give black precipitate with H_2S in presence of HCl solution.
 (3) Ferric ion gives blood red colour with potassium thiocyanate.
 (4) Cu^{2+} salts give red coloured borax bead test in reducing flame.

CQA044

17. The cation that will not be precipitated by H_2S in the presence of dil. HCl is: [JEE (Main) 2015]
 (1) Pb^{2+} (2) As^{3+} (3) Co^{2+} (4) Cu^{2+}

CQA036

18. When concentrated HCl is added to an aqueous solution of CoCl_2 , its colour changes from reddish pink to deep blue. Which complex ion gives blue colour in this reaction? [JEE (Main) 2015]

- (1) $[\text{CoCl}_4]^{2-}$ (2) $[\text{CoCl}_6]^{3-}$ (3) $[\text{CoCl}_6]^{4-}$ (4) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$

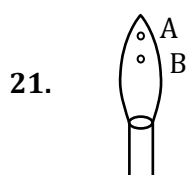
CQA038

19. Which of the following statements is incorrect? [JEE (Main) 2013]
- (1) Fe^{2+} ion also gives blood red colour with SCN^- ion
 - (2) Cupric ion reacts with excess of ammonia solution to give deep blue colour of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ion.
 - (3) Fe^{3+} ion gives blood red colour with SCN^- ion.
 - (4) On passing H_2S into Na_2ZnO_2 solution, a white ppt of ZnS is formed.

CQA033

20. Sodium Carbonate cannot be used in place of $(\text{NH}_4)_2\text{CO}_3$ for the identification of Ca^{2+} , Ba^{2+} and Sr^{2+} ions (in group V) during mixture analysis because : [JEE (Main) 2013]
- (1) Sodium ions will react with acid radicals
 - (2) Concentration of CO_3^{2-} ions is very low
 - (3) Mg^{2+} ions will also be precipitated
 - (4) Na^+ ions will interfere with the detection of Ca^{2+} , Ba^{2+} and Sr^{2+} ions

CQA032



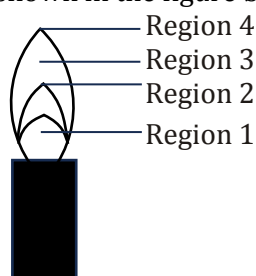
In a borax bead test under hot condition, a metal salt (one from the given) is heated at point B of the flame, resulted in green colour salt bead. The spin-only magnetic moment value of the salt is BM (Nearest integer)
[Given atomic number of Cu = 29, Ni = 28, Mn = 25, Fe = 26]

[JEE (Main) 2024]

CQA219

22. The hottest region of Bunsen flame shown in the figure below is :

[JEE (Main) 2016]



- (1) region 4 (2) region 1 (3) region 2 (4) region 3

CQA040

23. The metal ion whose electronic configuration is not affected by the nature of the ligand and which gives a violet colour in non-luminous flame under hot condition in borax bead test is

[JEE (Main) 2025]

- (1) Ti^{3+} (2) Ni^{2+} (3) Mn^{2+} (4) Cr^{3+}

CQA226

24. A pink coloured salt turns blue on heating. The presence of which cation is most likely ?

[JEE (Main) 2015]

- (1) Co^{2+} (2) Cu^{2+} (3) Zn^{2+} (4) Fe^{2+}

CQA039

EXERCISE - JEE (Advanced) PYQ

- In the following reaction sequence in aqueous solution, the species X, Y and Z, respectively, are

$$\text{S}_2\text{O}_3^{2-} \xrightarrow{\text{Ag}^+} \text{X} \xrightarrow{\text{Ag}^+} \text{Y} \xrightarrow{\text{with time}} \text{Z}$$

Clear solution white precipitate black precipitate

[JEE (Advanced) 2016]

(A) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$, $\text{Ag}_2\text{S}_2\text{O}_3$, Ag_2S (B) $[\text{Ag}(\text{S}_2\text{O}_3)_3]^{5-}$, Ag_2SO_3 , Ag_2S
 (C) $[\text{Ag}(\text{SO}_3)_2]^{3-}$, $\text{Ag}_2\text{S}_2\text{O}_3$, Ag (D) $[\text{Ag}(\text{SO}_3)_3]^{3-}$, Ag_2SO_4 , Ag

CQA076
- A mixture of two salts is used to prepare a solution S, which gives the following results :

White precipitate(s) only	$\xleftarrow[\text{Room temperature}]{\text{Dilute NaOH(aq.)}}$ (aq. solution of the salts)	$\xrightarrow[\text{Room temperature}]{\text{Dilute HCl(aq.)}}$ White precipitate(s) only
------------------------------	--	---

The correct option(s) for the salt mixture is(are) [JEE (Advanced) 2021]

(A) $\text{Pb}(\text{NO}_3)_2$ and $\text{Zn}(\text{NO}_3)_2$ (B) $\text{Pb}(\text{NO}_3)_2$ and $\text{Bi}(\text{NO}_3)_3$
 (C) AgNO_3 and $\text{Bi}(\text{NO}_3)_3$ (D) $\text{Pb}(\text{NO}_3)_2$ and $\text{Hg}(\text{NO}_3)_2$

CQA082
- With reference to aqua regia, choose the correct option(s). [JEE (Advanced) 2019]
- A colorless aqueous solution contains nitrates of two metals, X and Y. When it was added to an aqueous solution of NaCl, a white precipitate was formed. This precipitate was found to be partly soluble in hot water to give a residue P and a solution Q. The residue P was soluble in aq. NH_3 and also in excess sodium thiosulfate. The hot solution Q gave a yellow precipitate with KI. The metals X and Y, respectively, are [JEE (Advanced) 2020]

Paragraph for Question No. Q. 5 & 6

An aqueous solution of a mixture of two inorganic salts, when treated with dilute HCl, gave a precipitate (P) and a filtrate (Q). The precipitate P was found to dissolve in hot water. The filtrate (Q) remained unchanged, when treated with H_2S in a dilute mineral acid medium. However, it gave a precipitate (R) with H_2S in an ammoniacal medium. The precipitate R gave a coloured solution (S), when treated with H_2O_2 in an aqueous NaOH medium.

- The precipitate P contains [JEE (Advanced) 2013]
- The coloured solution S contains [JEE (Advanced) 2013]

Paragraph for Question No. Q.7 & 8

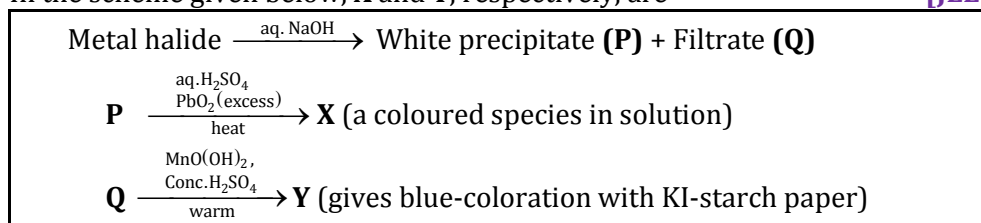
The reaction of $\text{K}_3[\text{Fe}(\text{CN})_6]$ with freshly prepared FeSO_4 solution produces a dark blue precipitate called Turnbull's blue. Reaction of $\text{K}_4[\text{Fe}(\text{CN})_6]$ with the FeSO_4 solution in complete absence of air produces a white precipitate X, which turns blue in air. Mixing the FeSO_4 solution with NaNO_3 , followed by a slow addition of concentrated H_2SO_4 through the side of the test tube produces a brown ring.

7. Precipitate X is
 (A) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (B) $\text{Fe}[\text{Fe}(\text{CN})_6]$ (C) $\text{K}_2\text{Fe}[\text{Fe}(\text{CN})_6]$ (D) $\text{KFe}[\text{Fe}(\text{CN})_6]$ [JEE (Advanced) 2021] CQA083
8. Among the following, the brown ring is due to the formation of
 (A) $[\text{Fe}(\text{NO})_2(\text{SO}_4)_2]^{2\ominus}$ (B) $[\text{Fe}(\text{NO})_2(\text{H}_2\text{O})_4]^{3\oplus}$ [JEE (Advanced) 2021] CQA084
 (C) $[\text{Fe}(\text{NO})_4(\text{SO}_4)_2]$ (D) $[\text{Fe}(\text{NO})(\text{H}_2\text{O})_5]^{3\oplus}$

Paragraph for Question No. 9 and 10

When potassium iodide is added to an aqueous solution of potassium ferricyanide, a reversible reaction is observed in which a complex **P** is formed. In a strong acidic medium, the equilibrium shifts completely towards **P**. Addition of zinc chloride to **P** in a slightly acidic medium results in a sparingly soluble complex **Q**.

9. The number of moles of potassium iodide required to produce two moles of **P** is _____. [JEE (Advanced) 2024] CQA220
10. The number of zinc ions present in the molecular formula of **Q** is _____. [JEE (Advanced) 2024] CQA221
11. In the scheme given below, **X** and **Y**, respectively, are [JEE (Advanced) 2023]



- (A) $\text{CrO}_4^{2\ominus}$ and Br_2 (B) $\text{MnO}_4^{2\ominus}$ and Cl_2 (C) $\text{MnO}_4^{\ominus}$ and Cl_2 (D) MnSO_4 and HOCl [JEE (Advanced) 2023] CQA222
12. The correct option(s) to distinguish nitrate salts of $\text{Mn}^{2\oplus}$ and $\text{Cu}^{2\oplus}$ taken separately is (are) [JEE (Advanced) 2018]
 (A) $\text{Mn}^{2\oplus}$ show the characteristic green colour in the flame test
 (B) Only $\text{Cu}^{2\oplus}$ show the formation of precipitate by passing H_2S in acidic medium
 (C) Only $\text{Mn}^{2\oplus}$ show the formation of precipitate by passing H_2S in faintly basic medium
 (D) $\text{Cu}^{2\oplus}/\text{Cu}$ has higher reduction potential than $\text{Mn}^{2\oplus}/\text{Mn}$ (measured under similar conditions) CQA077
13. Among PbS , CuS , HgS , MnS , Ag_2S , NiS , CoS , Bi_2S_3 and SnS_2 , the total number of BLACK coloured sulphides is : [JEE (Advanced) 2014] CQA074
14. Upon treatment with ammoniacal H_2S , the metal ion that precipitates as a sulphide is : [JEE (Advanced) 2013]
 (A) Fe(III) (B) Al (III) (C) Mg(II) (D) Zn(II) CQA071
15. The pair(s) of ions where BOTH the ions are precipitated upon passing H_2S gas in presence of dilute HCl , is(are) [JEE (Advanced) 2015]
 (A) $\text{Ba}^{2\oplus}$, $\text{Zn}^{2\oplus}$ (B) $\text{Bi}^{3\oplus}$, $\text{Fe}^{3\oplus}$ (C) $\text{Cu}^{2\oplus}$, $\text{Pb}^{2\oplus}$ (D) $\text{Hg}^{2\oplus}$, $\text{Bi}^{3\oplus}$ CQA075
16. The green colour produced in the borax bead test of a chromium(III) salt is due to - [JEE (Advanced) 2019]
 (A) $\text{Cr(BO}_2)_3$ (B) CrB (C) $\text{Cr}_2(\text{B}_4\text{O}_7)_3$ (D) Cr_2O_3 CQA078
17. Thermal decomposition of AgNO_3 produces two paramagnetic gases. The total number of electrons present in the antibonding molecular orbitals of the gas that has the higher number of unpaired electrons is _____. [JEE (Advanced) 2022] CQA085

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	A	A	C	B	A	A	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	B	C	C	D	A	B	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	C	B	D	C	B	B	D	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	B	C	A	C	D	D	B	B
Que.	41	42	43	44	45	46				
Ans.	A	B	D	C	B	B				

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABC	AD	ABCD	ABC	BC	BCD	BCD	AC	AD	AD
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	AB	AB	ACD	ACD	BD	B	D	A	D	B
Que.	21	22	23	24	25	26				
Ans.	C	D	B	C	D	C				

EXERCISE (S)

1.	30	2.	0	3.	3	4.	7	5.	3
6.	7	7.	3	8.	2	9.	40	10.	2
11.	1	12.	4	13.	2	14.	6	15.	8
16.	0.33	17.	3.874						

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	1	4	4	4	3	4	2	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	2	4	1	2	3	1	1	3
Que.	21	22	23	24						
Ans.	6	3	2	1						

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	AB	BCD	A	A	D	C	D	2	3 or 2
Que.	11	12	13	14	15	16	17			
Ans.	C	BD	7	D	CD	A	6			